

Q1. Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

FEATURE	DDR3	DDR4	DDR5
Release Year	2007	2014	2020
Speed	800-2133 MT/s	1600-3200 MT/s	4800-8400+ MT/s
Voltage	1.5 V	1.2 V	1.1 V
Power Use	High	Medium	Low
Maxi. Capa.	Up to 16 GB	Up to 64	Up to 128
Gaming	Good for old games	Good for modern games	Good for latest games
Multitasking	Basic	Smooth	Very smooth
Performance	Good	Better	Best
Price	Cheapest	Medium	Most Expensive
Typical Use	Old PCs, office work	Home, office, gaming	High-end gaming, AI, video editing, workstations

Q2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.



1. Turn off the computer or laptop.
2. Unplug the power cable and charger.
3. Open the back cover (laptop) or side panel (desktop).
4. Find the RAM slot on the motherboard.
5. Push the two side clips outward. The RAM will pop up.
6. Remove the RAM carefully by holding the edges.
7. Take a photo of the RAM slot and RAM module.
8. Insert the RAM back into the same slot.
9. Press gently until the clips lock automatically.
10. Close the cover and reconnect the power.
11. Turn on the computer.
12. Check that the computer starts normally. If it boots successfully, the RAM is installed correctly.

Q3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Storage Type	Speed	Best Feature	Example App / Use Case
HDD	Slow	Cheap and large storage	Movies, Photos, Videos
SSD	Fast	Faster than HDD	Microsoft Word, Google Chrome
NVMe SSD	Very Fast	Fastest storage	Games (PUBG, GTA V), Video Editing

1 HDD (HARD DISK DRIVE) :

- ✓ Slow speed, low price, good for storing large files.

Use: Movies, photos, and videos.

2 SSD (SATA SOLID STATE DRIVE) :

- ✓ Faster than HDD, starts the computer quickly, more reliable.

Use: Microsoft Word and Google Chrome.

3 NVMe SSD (FASTEST STORAGE) :

- ✓ Fastest storage, loads apps and games very quickly.

Use: PUBG, GTA V, or video editing software.

Q4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

